

EDUCATION

University of California, LA (UCLA)	M.S. Aerospace Engineering	09/2026 – 06/2028
San Diego State University (SDSU)	B.S. Aerospace Engineering (Minor in Physics)	08/2023 – 05/2026
Santa Monica College (SMC)	A.A. General Science (with Honors)	06/2020 – 07/2023

EXPERIENCE

Instructional Student Assistant **SDSU College of Engineering** **01/2025 – 12/2025**

- Provide academic support for AE320 (Astrodynamics), on topics like two-body orbital mechanics on Keplerian orbits and orbital transfers.
- Provide academic support for AE280 (Differential Equations), reinforcing topics such as Laplace transforms, Fourier series, and linear algebra in engineering contexts through tutoring, preparation, and coordination with faculty.

Undergraduate Research Assistant (GNC) **SDSU SPACE Lab** **11/2023 – 08/2024**

- Programmed DJI Tello EDU drones using djitellopy (Python) for autonomous QR code navigation and string-based command execution.
- Applied OpenCV to extract and decode QR codes, then estimated drone-to-code distance with a pixel-scaling method (± 0.5 in accuracy).
- Designed QR-coded indoor navigation paths; drone detected, approached, and followed sequences of embedded movement commands.
- Integrated machine learning-based landing pad detection and tested basic object avoidance using monocular SLAM (ORB-SLAM3).
- Adapted lab environment by applying textured surfaces to improve LiDAR-based altitude control in low-feature terrain.

Avionics and Propulsion Engineer **SDSU Rocket Project** **08/2023 – 07/2024**

- Designed and 3D printed custom mounts using SOLIDWORKS for Pressure Transducers, Thermocouples, and Solenoids interfacing with National Instruments DAQ for static fire test data acquisition.
- Assembled Avionics Ground System by soldering and wiring sensors to NI DAQ and developing bulkhead panels for sensor and power interfaces.
- Developed thermal-resistant hardware enclosures for electronics and interface modules using ASA/PLA filaments on Bambu Lab 3D printers.
- Researched optimal inlet-to-throat ratios for ethanol-fed Venturi flow meters in Electric Turbopump testing; drafted Arduino code to measure pressure differentials with T200 series sensors before project shift.

Recreation Aide **City of Beverly Hills** **04/2019 – 07/2023**

- Led structured activities for children ages 4–14 in camp programs: sports, educational games, arts and crafts, and academic support.
- Adapted during COVID19 to develop remote programming and assisted with recreation operations, including tennis court management.

PUBLICATIONS

Khodadi, P., Cook, A., Tobita, Y., Liu, X., "AIAA SDSU Student Branch History," *AIAA SciTech 2026*, AIAA paper 2026-2727, Jan. 2026.

PROGRAMS

Mission Concept Academy (NASA L'SPACE) **05/2022 – 09/2022**

- Calculated net heat transfer across rover systems and designed thermally optimized housing for Martian rock samples and electronics.
- Selected materials for passive thermal control and developed a dust-proof thermal dissipation panel based on mission constraints.
- Done modules in Siemens NX, Risk Management, Project Management, Systems Engineering, and Heat Transfer.
- Collaborated with a 12-member interdisciplinary team to design a Mars rover mission targeting lava tube exploration.

NASA Proposal Writing & Evaluation Experience (L'SPACE)

01/2022 – 05/2022

- Developed a manufacturing and prototyping workflow for 2DPA-1 polymer (MIT-discovered material) in space applications.
- Built a year-long Gantt-style timeline to synchronize engineering, research, and testing phases into a cohesive project schedule.
- Participated in NASA-style proposal evaluation, applying official scoring criteria to competing project submissions.
- Supported proposal design using 2DPA-1 in space applications, submitting to NASA for funding consideration.

NASA AFRC Engineering Design Challenge (NCAS Mission 3)

07/2022

- Led the Engineering sub-team during a NASA Armstrong center program developing wheelchair-accessible seating for eVTOL systems.
- Designed a fold-flat seating mechanism to enable autonomous docking for wheelchairs, integrating mechanical and automation features.
- Shadowed NASA professionals—including a principal investigator and a Pathways intern—and toured advanced eVTOL research facilities.

SCHOOL PROJECTS

A E 403W – Senior Project | Presented at SDSU Senior Design 2026

- Designing autonomous collision avoidance for NASA Baseball Avoidance Multirotor (BAM) challenge.
- Building the MATLAB / Simulink simulation for the drone's collision avoidance with a baseball.
- Testing the success rate of the avoidance program with varying levels of sensor noise.
- Role: Project Lead – Responsible for the end-to-end success of the perception, prediction, and planning modules.

A E 460A + 460B – Spacecraft Design | Presented at SDSU Senior Design 2026

- Developing the conceptual and preliminary design of a Solar Gravitational Lens spacecraft mission: covering mission definition, payload integration, systems architecture, and community-impact considerations.
- Role: Flight Dynamics Systems Engineer – Planning the mission architecture and flight trajectory of the Spacecraft.
- Designing the entire Mission Control Sequence within AGI Systems Tool Kit's Astrogator.

PERSONAL PROJECTS

Atmospheric Magnetoplasmadynamic (MPD) Ion Thruster built using a 1s Lipo battery, DC transformer, nickel strips, and an empty soda can.

YouTube channel "SciPK" where I explain science and engineering topics through video essays.

EXTRACURRICULAR ACTIVITIES

- As **Outreach Officer** of **American Institute of Aeronautics and Astronautics SDSU branch** I organize and lead company tours, speaker panels, and museum events to connect aerospace Engineering students with industry professionals. I also wrote the history paper found in the publications section.
- **Membership Officer** of **Persian Student Association** at SDSU.

AWARDS

- 2025-2026 Reuben H. Fleet Scholarship – AIAA San Diego
- 2026 Outstanding Contribution to the AIAA Student Chapter at SDSU

SKILLS

<u>Software</u>	<u>Hardware</u>	<u>Other Technical or Soft-Skills</u>
• C/C++ • Python • OpenCV • Robot Operating System (ROS) • Colosseum / AirSim • CAD (SOLIDWORKS and NX) • MATLAB / Simulink • AGI Systems Tool Kit (STK) • STK Astrogator • JMARS • Version Control (Git) • Adobe Premiere Pro • Unreal Engine	• Data Acquisition (DAQ) • Soldering • Wiring • 3D Printing	• Public Speaking • LaTeX (via Overleaf) • Problem-Solving & Critical Thinking • Teamwork & Collaboration • Adaptability • Attention to Detail • Proposal Evaluation • Mission Concept Design • Control System Design • Orbital mechanics • Research - General • Bilingual: Persian, English